

NAME: ADOYI NATHANIEL JOHNSON

STUDENT ID: FWSD2511203

STUDENT EMAIL: fwsd2511203@students.icdfa.edu.ng

COURSE CODE: SBT-DF203-BASIC NETWORKING SKILLS

LAB NAME: LAB7 DNS INTRODUCTION and TRAFFIC ANALYSIS

DATE: 22/09/26

INSTRUCTOR: AMINU IDRIS

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 7: DNS Introduction and Traffic Analysis

Three-Week Delivery Block 3/3 | 19-25 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 7
Lab Title	DNS Introduction and Traffic Analysis
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	dig_dns.pcap or locally generated DNS captures; optional smtp.pcap correlation

1. Source Materials and Slide Alignment

- Domain Name System Forensics: domain-to-IP resolution, hierarchical DNS, resolver/TLD/authoritative roles, dig queries, capture with tshark, query/response fields and DNS within email traffic.
- The slide practical captures dig and browser DNS traffic, examines A records, local resolver configuration, TTL and authoritative information.
- The exercise builds the normal DNS baseline required before analyzing spoofing in Lab 8.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

You are documenting normal DNS behaviour for an incident-response baseline. You will identify the configured resolver, capture controlled dig queries, analyze transaction IDs, names, record types, response codes, answers and TTL values, and correlate DNS resolution with a subsequent web or email connection.

3. Learning Outcomes

- Explain recursive and iterative DNS resolution at a foundational level.
- Use dig to query A, AAAA, MX and NS records.
- Capture and analyze DNS queries and responses with tshark/Wireshark.
- Identify transaction ID, query name, record type, response code, answer and TTL.
- Correlate DNS responses with later IP connections.
- Recognize normal variations such as multiple answers, CNAMEs, cache responses and IPv6 queries.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.

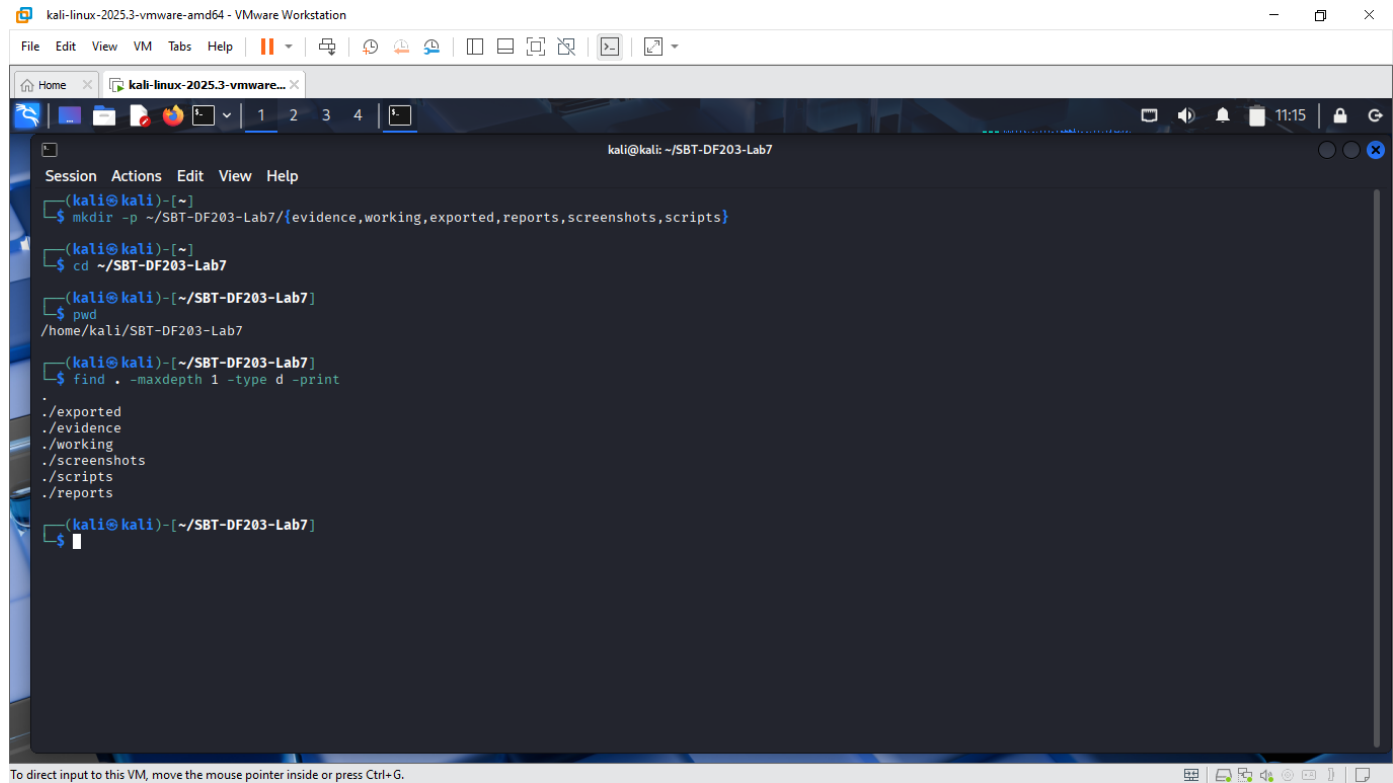
5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Generate and analyze DNS traffic.
DNS utility	dig from dnsutils	Perform controlled queries.
Capture tools	tshark and Wireshark	Capture DNS traffic.
Evidence	dig_dns.pcap or new PCAPNG	Analysis source.
Resolver config	/etc/resolv.conf, resolvectl	Identify configured DNS service.
Optional browser	Firefox/Chrome	Generate multiple DNS lookups.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab7
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working

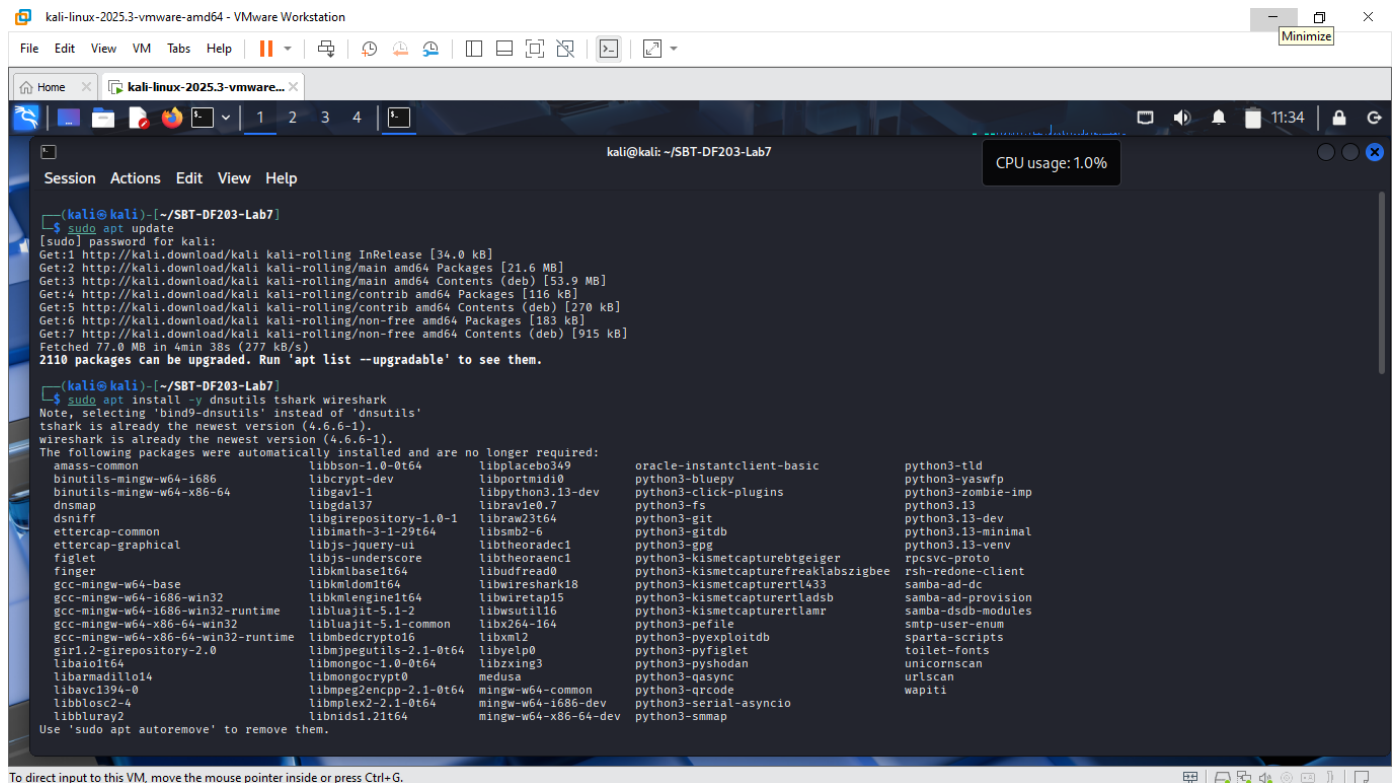
ANSWER:

The screenshot shows a terminal window titled 'kali@kali: ~/SBT-DF203-Lab7'. The terminal displays the following commands and their outputs:

```
(kali@kali)-[~]  
$ mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}  
(kali@kali)-[~]  
$ cd ~/SBT-DF203-Lab7  
(kali@kali)-[~/SBT-DF203-Lab7]  
$ pwd  
/home/kali/SBT-DF203-Lab7  
(kali@kali)-[~/SBT-DF203-Lab7]  
$ find . -maxdepth 1 -type d -print  
.  
./exported  
./evidence  
./working  
./screenshots  
./scripts  
./reports  
(kali@kali)-[~/SBT-DF203-Lab7]  
$
```

```
sudo apt update  
sudo apt install -y dnsutils tshark wireshark  
cat /etc/resolv.conf | tee reports/resolv_conf.txt  
resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true  
wget -O evidence/dig_dns.pcap \  
  'https://raw.githubusercontent.com/frankwxu/digital-forensics-  
  lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'  
cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap  
sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
```

Screenshot required: Configured resolver information, evidence details and hashes.

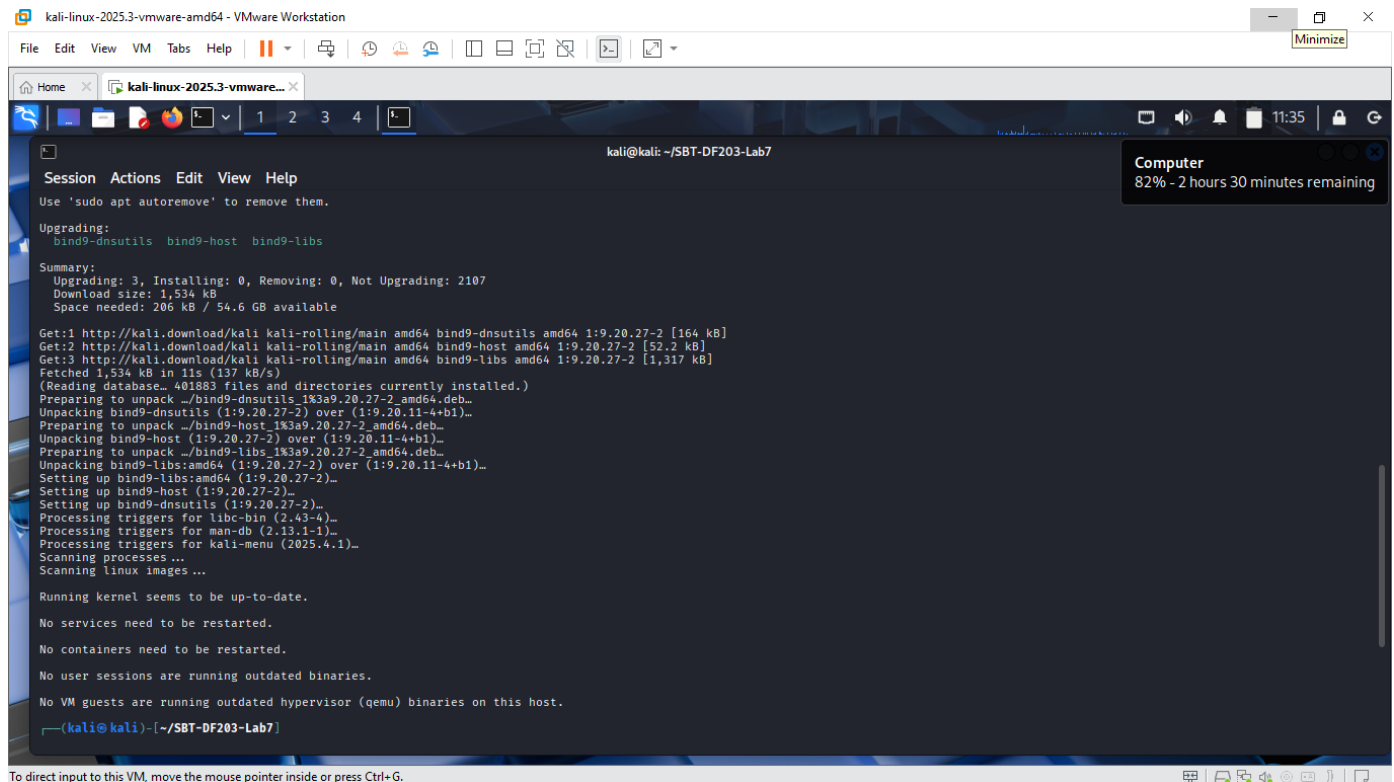


```
kali@kali: ~/SBT-DF203-Lab7
$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.9 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 77.0 MB in 4min 38s (277 kB/s)
2110 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)~/SBT-DF203-Lab7
$ sudo apt install -y dnstools tshark wireshark
Note, selecting 'bind9-dnstools' instead of 'dnstools'
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  amass-common libbson-1.0-0t64 libplacebo349 oracle-instantclient-basic
  binutils-mingw-w64-1686 libcrypto-dev libportmidi0 python3-bluepy
  binutils-mingw-w64-x86-64 libgavl-1 libpython3.13-dev python3-click-plugins
  dnsmasq libgdal37 libravie0.7 python3-fs
  dsntiff libgirepository-1.0-1 libraw23t64 python3-git
  ettercap-common libamath-3-1-29t64 libsm2-6 python3-pitdb
  ettercap-graphical libjs-jquery-ui libtheoraec1 python3-gpg
  figlet libjs-underscore libtheoraenc1 python3-kismetcapturebtgeiger
  finger libkmlbase1t64 libudfread0 python3-kismetcapturefreaklszigbee
  gcc-mingw-w64-base libkmlengine1t64 libwireshark18 python3-kismetcapturertl433
  gcc-mingw-w64-1686-win32 libkmlengine1t64 libwiretap15 python3-kismetcapturetrtladsb
  gcc-mingw-w64-1686-win32-runtime liblua5.1-2 libwsutil16 python3-kismetcapturetrtlamr
  gcc-mingw-w64-x86-64-win32 liblua5.1-2-common libx264-164 python3-pefile
  gcc-mingw-w64-x86-64-win32-runtime libmbcdecrypt0 libx265 python3-pyexploitdb
  gir1.2-girepository-2.0 libmpegutils-2.1-0t64 libyelp0 python3-pyfiglet
  libaio1t64 libmongoc-1.0-0t64 libzxing3 python3-pyshodan
  libarmadillo14 libmongocrypt0 medusa python3-qasync python3-qrcode
  libavc1394-0 libmpeg2encpp-2.1-0t64 mingw-w64-1686-dev python3-serial-asyncio
  libblosc2-4 libmplex2-2.1-0t64 mingw-w64-1686-dev python3-smmap
  libbluray2 libnids1.21t64 mingw-w64-x86-64-dev python3-smmmap

Use 'sudo apt autoremove' to remove them.
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali@kali: ~/SBT-DF203-Lab7
$ sudo apt autoremove
Use 'sudo apt autoremove' to remove them.

Upgrading:
  bind9-dnstools bind9-host bind9-lbns

Summary:
  Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 2107
  Download size: 1,534 kB
  Space needed: 206 kB / 54.6 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 bind9-dnstools amd64 1:9.20.27-2 [164 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 bind9-host amd64 1:9.20.27-2 [52.2 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 bind9-lbns amd64 1:9.20.27-2 [1,317 kB]
Fetched 1,534 kB in 11s (137 kB/s)
(Reading database... 401883 files and directories currently installed.)
Preparing to unpack ./bind9-dnstools_1k3a9.20.27-2_amd64.deb...
Unpacking bind9-dnstools (1:9.20.27-2) over (1:9.20.11-4+b1)...
Preparing to unpack ./bind9-host_1k3a9.20.27-2_amd64.deb...
Unpacking bind9-host (1:9.20.27-2) over (1:9.20.11-4+b1)...
Preparing to unpack ./bind9-lbns_1k3a9.20.27-2_amd64.deb...
Unpacking bind9-lbns:amd64 (1:9.20.27-2) over (1:9.20.11-4+b1)...
Setting up bind9-lbns:amd64 (1:9.20.27-2)...
Setting up bind9-host (1:9.20.27-2)...
Setting up bind9-dnstools (1:9.20.27-2)...
Processing triggers for libc-bin (2.43-4)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2025.4.1)...
Scanning processes ...
Scanning linux images ...

Running kernel seems to be up-to-date.

No services need to be restarted.

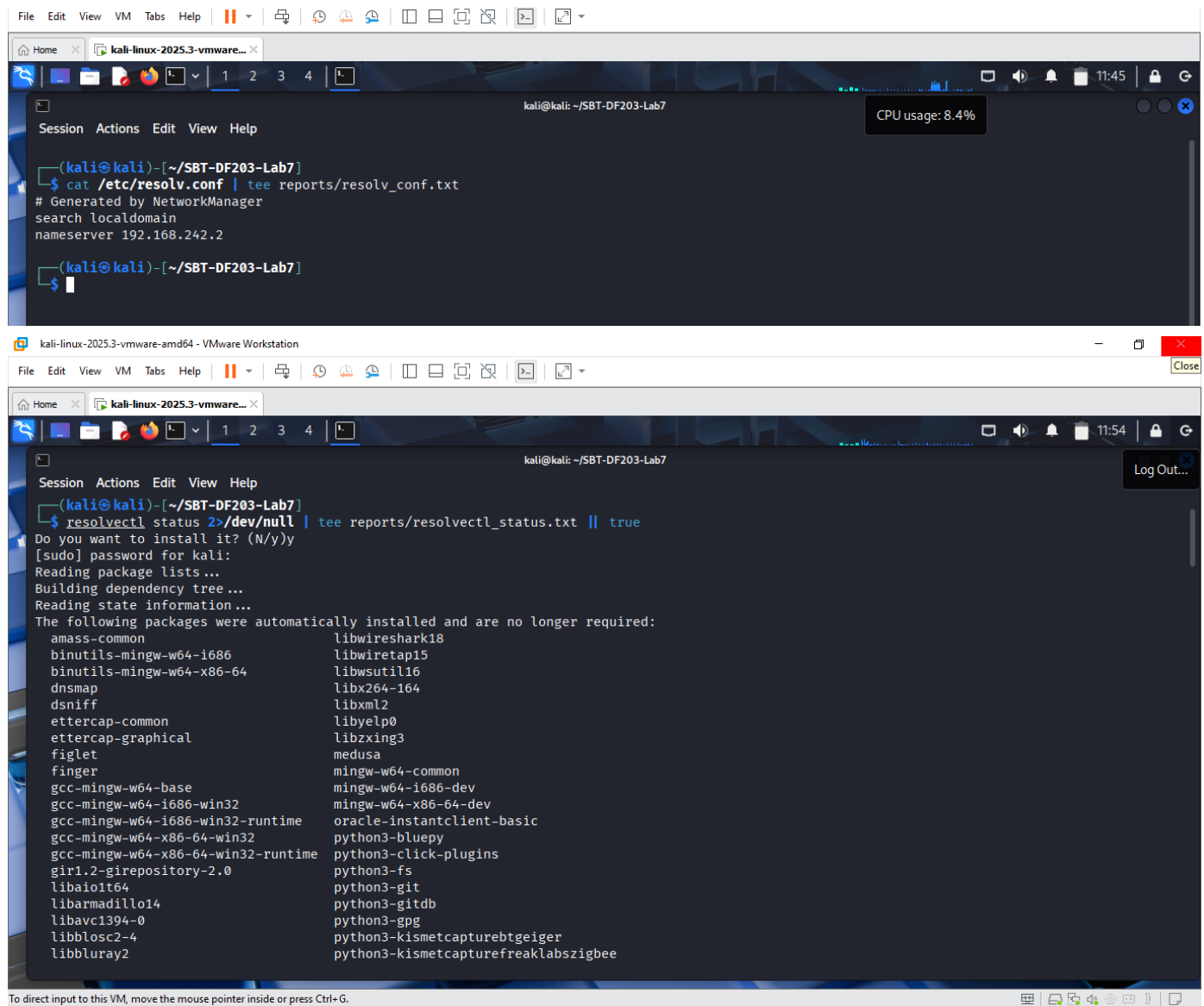
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

(kali@kali)~/SBT-DF203-Lab7
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali@kali: ~/SBT-DF203-Lab7
CPU usage: 8.4%

(kali@kali)~/SBT-DF203-Lab7
$ cat /etc/resolv.conf | tee reports/resolv_conf.txt
# Generated by NetworkManager
search localdomain
nameserver 192.168.242.2

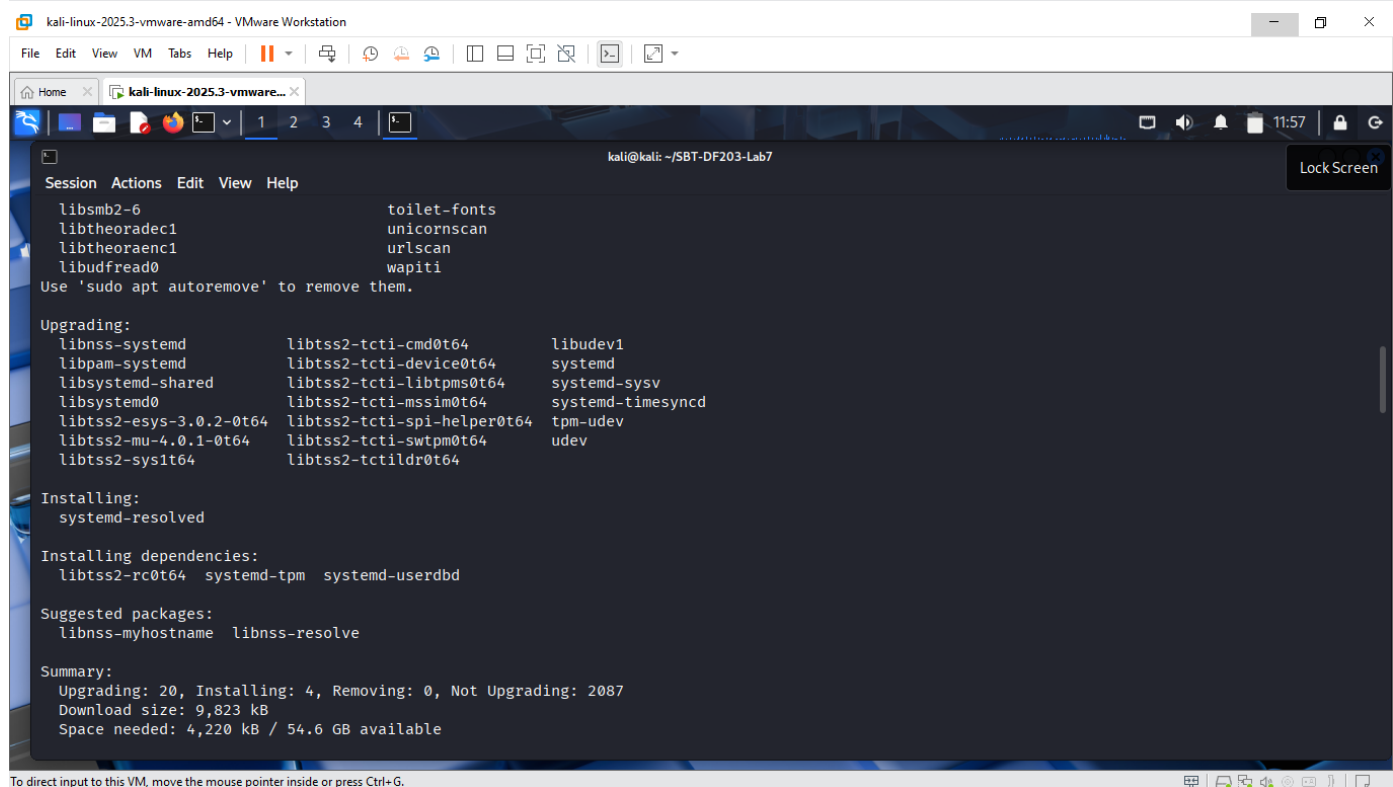
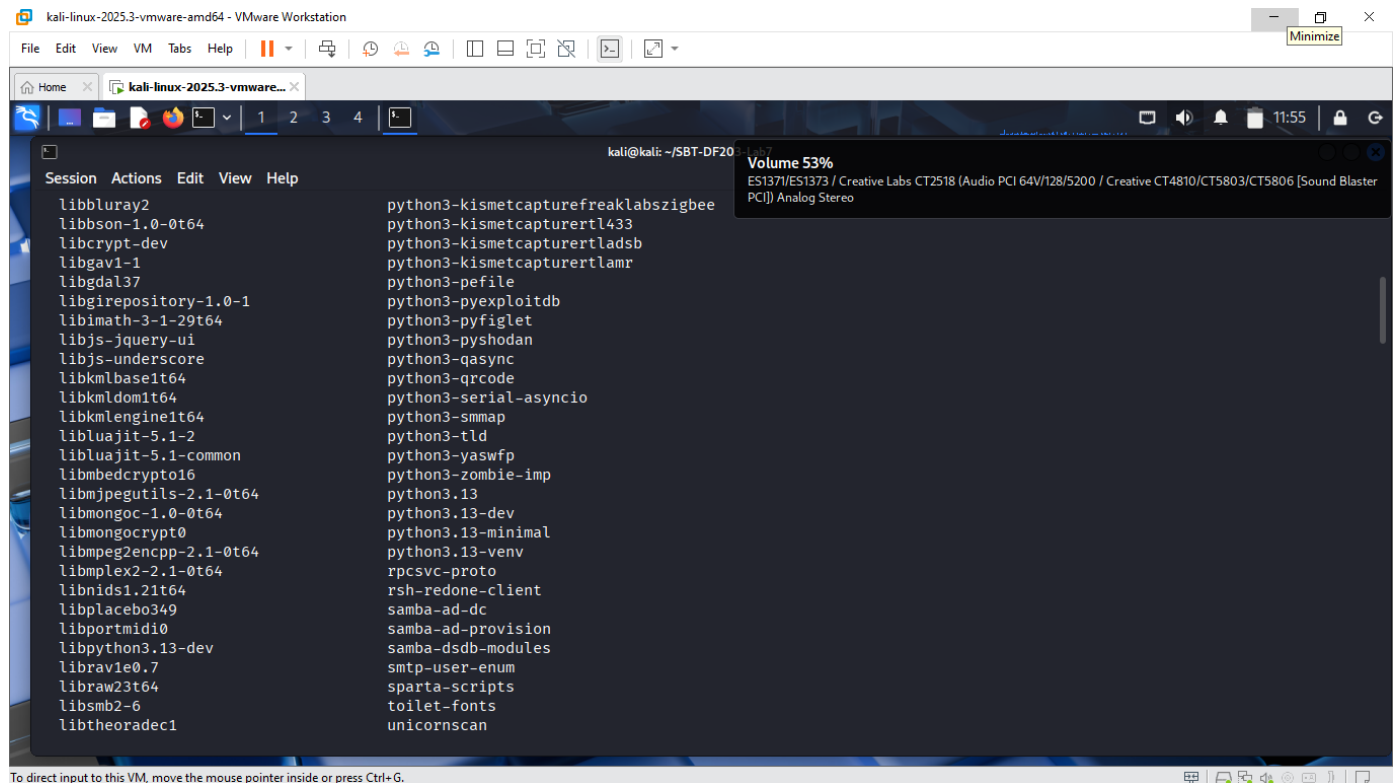
(kali@kali)~/SBT-DF203-Lab7
$

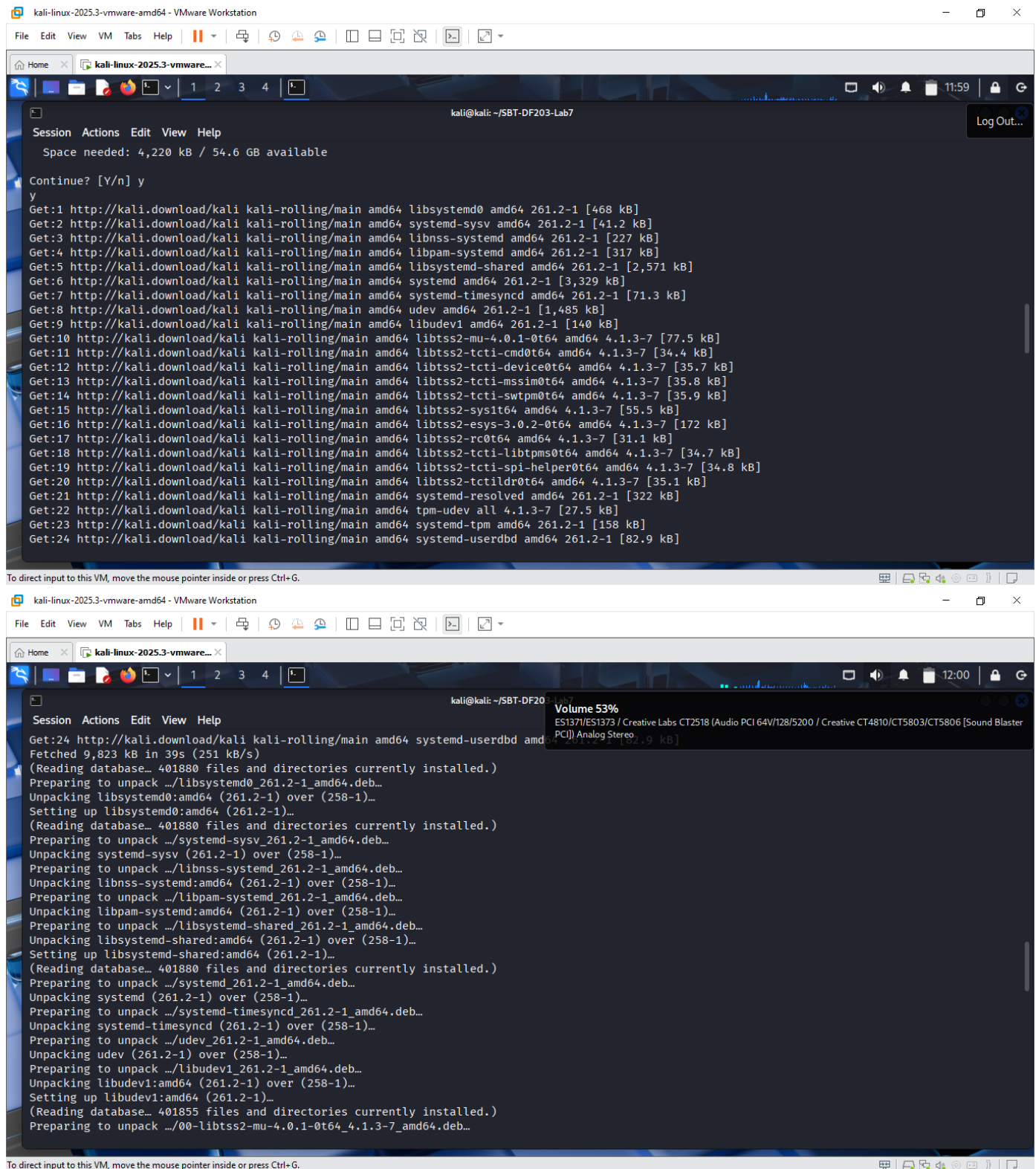
kali-linux-2025.3-vmware-amd64 - VMware Workstation
Close

(kali@kali)~/SBT-DF203-Lab7
Log Out...

(kali@kali)~/SBT-DF203-Lab7
$ resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
Do you want to install it? (N/y)y
[sudo] password for kali:
Reading package lists...
Building dependency tree...
Reading state information...
The following packages were automatically installed and are no longer required:
  amass-common          libwireshark18
  binutils-mingw-w64-i686  libwiretap15
  binutils-mingw-w64-x86_64  libwsutil16
  dnsmap                libx264-164
  dsniff                 libxml2
  ettercap-common        libyelp0
  ettercap-graphical     libzxing3
  figlet                 medusa
  finger                 mingw-w64-common
  gcc-mingw-w64-base      mingw-w64-i686-dev
  gcc-mingw-w64-i686-win32  mingw-w64-x86_64-dev
  gcc-mingw-w64-i686-win32-runtime  oracle-instantclient-basic
  gcc-mingw-w64-x86_64-win32  python3-bluepy
  gcc-mingw-w64-x86_64-win32-runtime  python3-click-plugins
  gir1.2-girepository-2.0  python3-fs
  libaio1t64             python3-git
  libarmadillo14          python3-gitdb
  libavc1394-0            python3-gpg
  libblsc2-4              python3-kismetcapturebtgeiger
  libbluray2              python3-kismetcapturefreaklabszigbee
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

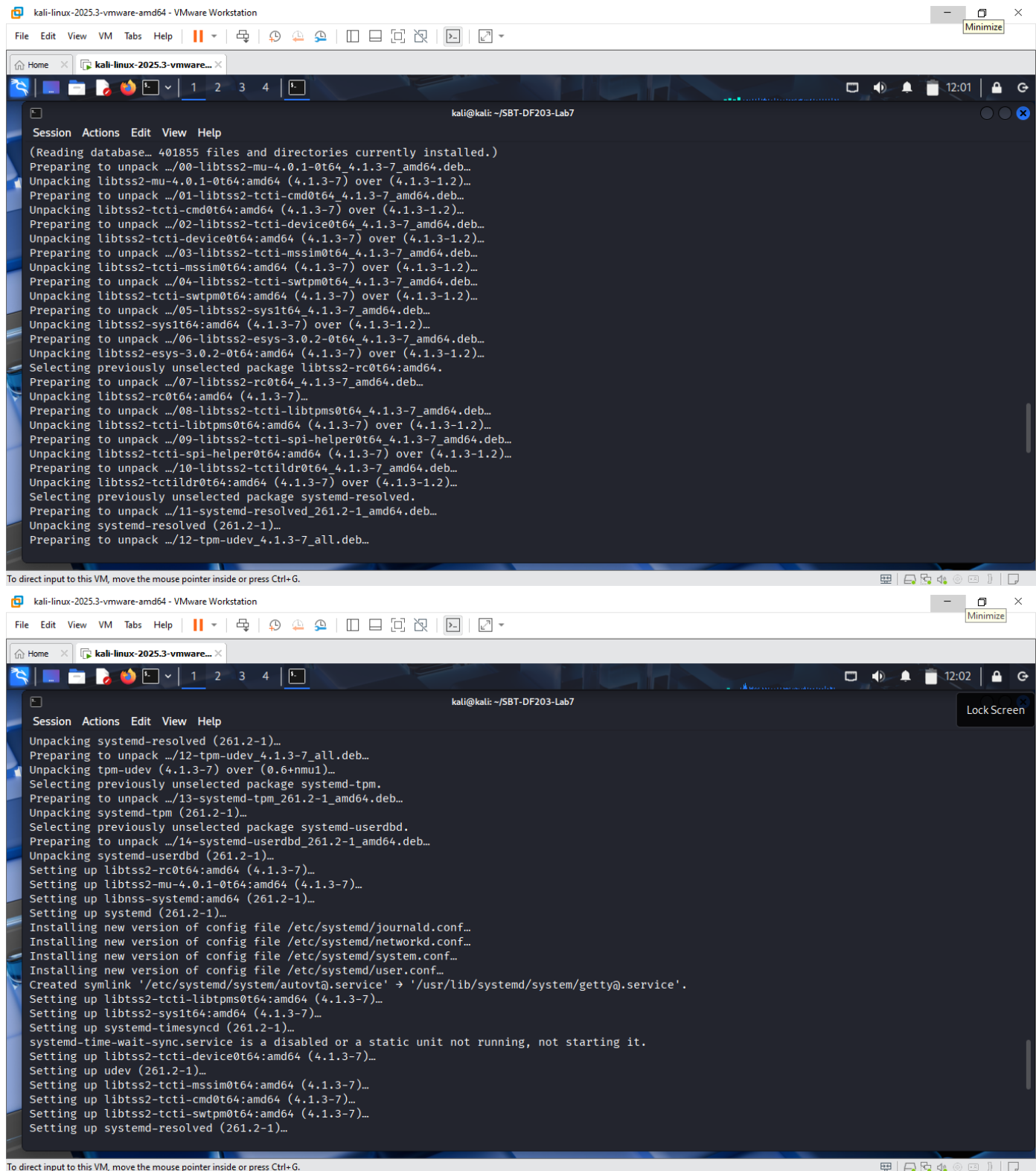




The image consists of two screenshots of a Kali Linux virtual machine (VM) running in VMware Workstation. The VM is named 'kali-linux-2025.3-vmware-amd64'.

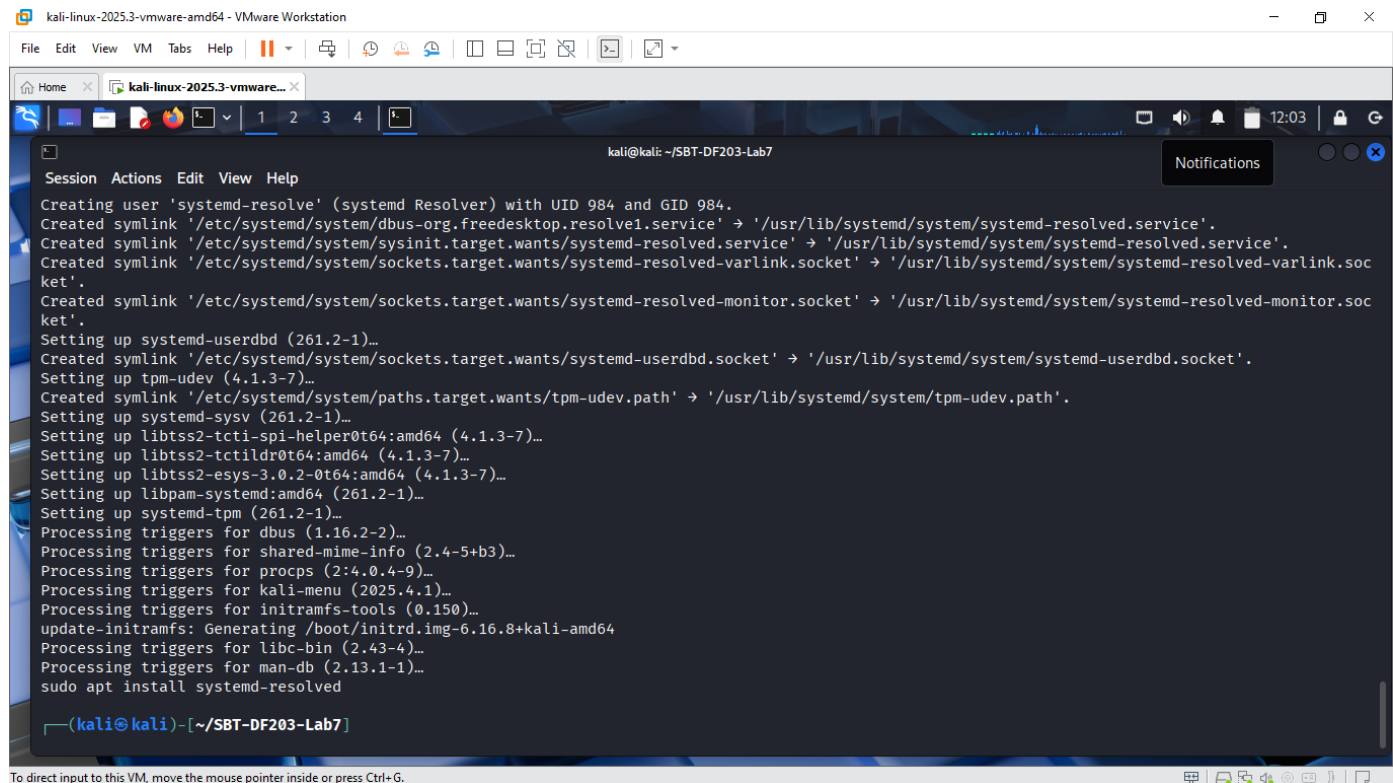
Top Screenshot: The terminal window shows the user at the prompt 'kali@kali: ~/SBT-DF203-Lab7'. The user has entered 'y' to continue the installation. The terminal displays a list of packages being downloaded from the Kali Rolling repository, including libsystemd0, systemd-sysv, libnss-systemd, libpam-systemd, libsystemd-shared, systemd, systemd-timesyncd, udev, libudev1, libtss2-mu-4.0.1-0t64, libtss2-tcti-cmd0t64, libtss2-tcti-device0t64, libtss2-tcti-mssim0t64, libtss2-tcti-swtpm0t64, libtss2-sysit64, libtss2-esys-3.0.2-0t64, libtss2-rc0t64, libtss2-tcti-libtpms0t64, libtss2-tcti-spi-helper0t64, libtss2-tctildr0t64, systemd-resolved, tpm-udev, and systemd-tpm. The download progress is shown for each package.

Bottom Screenshot: The terminal window shows the user at the prompt 'kali@kali: ~/SBT-DF203-Lab7'. The user has entered 'y' to continue the installation. The terminal displays the progress of the installation, showing the packages being unpacked and the system being set up. The packages being unpacked include libsystemd0, systemd-sysv, libnss-systemd, libpam-systemd, libsystemd-shared, systemd, systemd-timesyncd, udev, libudev1, libtss2-mu-4.0.1-0t64, libtss2-tcti-cmd0t64, libtss2-tcti-device0t64, libtss2-tcti-mssim0t64, libtss2-tcti-swtpm0t64, libtss2-sysit64, libtss2-esys-3.0.2-0t64, libtss2-rc0t64, libtss2-tcti-libtpms0t64, libtss2-tcti-spi-helper0t64, libtss2-tctildr0t64, systemd-resolved, tpm-udev, and systemd-tpm. The installation progress is shown for each package.



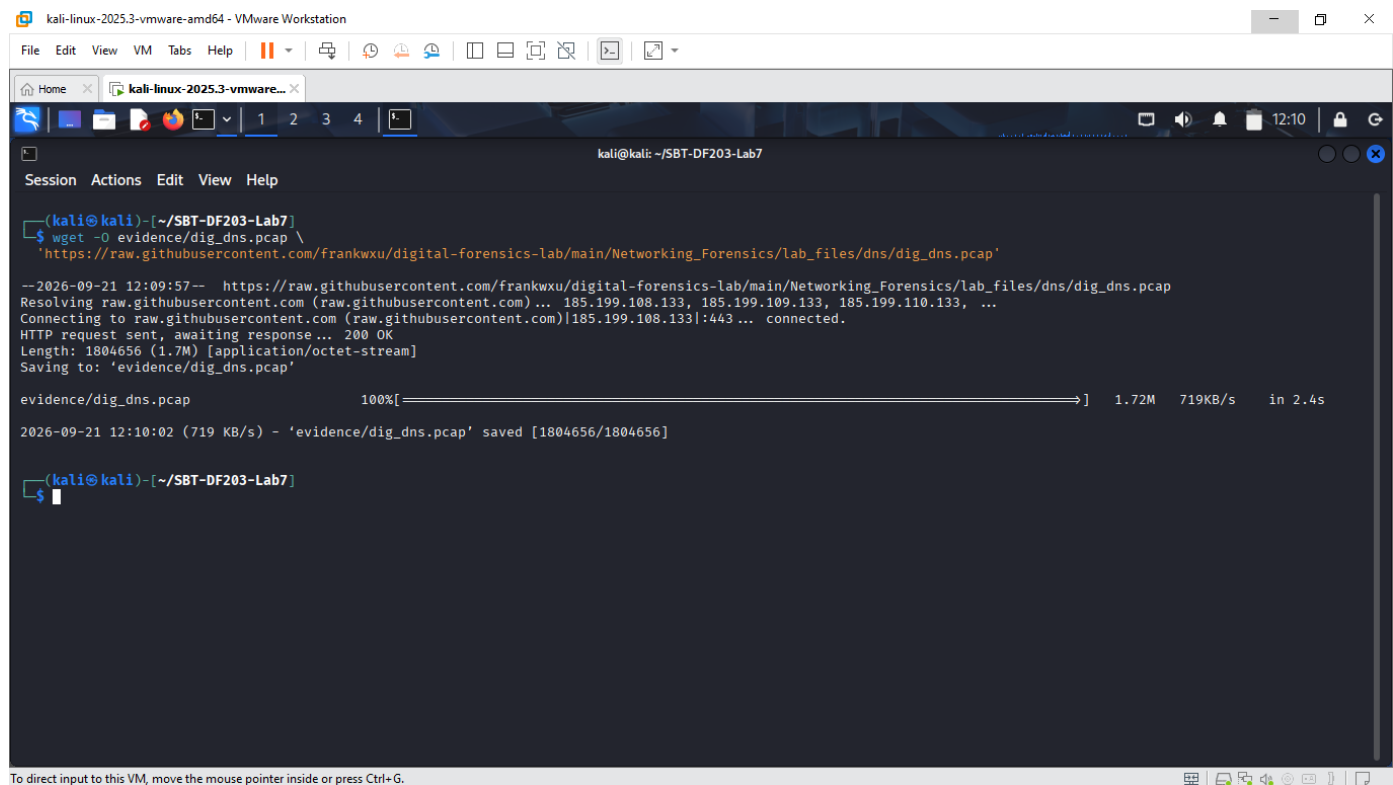
```
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
(Reading database... 401855 files and directories currently installed.)
Preparing to unpack .../00-libtss2-mu-4.0.1-0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-mu-4.0.1-0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../01-libtss2-tcti-cmd0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-tcti-cmd0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../02-libtss2-tcti-device0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-tcti-device0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../03-libtss2-tcti-mssim0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-tcti-mssim0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../04-libtss2-tcti-swtpm0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-tcti-swtpm0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../05-libtss2-sys1t64_4.1.3-7_amd64.deb...
Unpacking libtss2-sys1t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../06-libtss2-esys-3.0.2-0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-esys-3.0.2-0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Selecting previously unselected package libtss2-rc0t64:amd64.
Preparing to unpack .../07-libtss2-rc0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-rc0t64:amd64 (4.1.3-7)...
Preparing to unpack .../08-libtss2-tcti-libtpms0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-tcti-libtpms0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../09-libtss2-tcti-spi-helper0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-tcti-spi-helper0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Preparing to unpack .../10-libtss2-tctildr0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-tctildr0t64:amd64 (4.1.3-7) over (4.1.3-1.2)...
Selecting previously unselected package systemd-resolved.
Preparing to unpack .../11-systemd-resolved_261.2-1_amd64.deb...
Unpacking systemd-resolved (261.2-1)...
Preparing to unpack .../12-tpm-udev_4.1.3-7_all.deb...

kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
Unpacking systemd-resolved (261.2-1)...
Preparing to unpack .../12-tpm-udev_4.1.3-7_all.deb...
Unpacking tpm-udev (4.1.3-7) over (0.6+nmui)...
Selecting previously unselected package systemd-tpm.
Preparing to unpack .../13-systemd-tpm_261.2-1_amd64.deb...
Unpacking systemd-tpm (261.2-1)...
Selecting previously unselected package systemd-userdbd.
Preparing to unpack .../14-systemd-userdbd_261.2-1_amd64.deb...
Unpacking systemd-userdbd (261.2-1)...
Setting up libtss2-rc0t64:amd64 (4.1.3-7)...
Setting up libtss2-mu-4.0.1-0t64:amd64 (4.1.3-7)...
Setting up libtss2-sys1t64:amd64 (4.1.3-7)...
Setting up systemd (261.2-1)...
Installing new version of config file /etc/systemd/journald.conf...
Installing new version of config file /etc/systemd/networkd.conf...
Installing new version of config file /etc/systemd/system.conf...
Installing new version of config file /etc/systemd/user.conf...
Created symlink '/etc/systemd/system/autovt@.service' → '/usr/lib/systemd/system/getty@.service'.
Setting up libtss2-tcti-libtpms0t64:amd64 (4.1.3-7)...
Setting up libtss2-tcti-device0t64:amd64 (4.1.3-7)...
Setting up systemd-timesyncd (261.2-1)...
systemd-time-wait-sync.service is a disabled or a static unit not running, not starting it.
Setting up libtss2-tcti-swtpm0t64:amd64 (4.1.3-7)...
Setting up udev (261.2-1)...
Setting up libtss2-tcti-mssim0t64:amd64 (4.1.3-7)...
Setting up libtss2-tcti-cmd0t64:amd64 (4.1.3-7)...
Setting up libtss2-tcti-swtpm0t64:amd64 (4.1.3-7)...
Setting up systemd-resolved (261.2-1)...
```



The screenshot shows a Kali Linux terminal window titled "kali@kali: ~/SBT-DF203-Lab7". The terminal displays the output of the command `sudo apt install systemd-resolved`. The output lists various system components being configured, including the creation of the `systemd-resolve` user, symlinks for `systemd-resolved.service` and `systemd-resolved-varlink.socket`, and the setup of `systemd-userdbd`, `tpm-udev`, `systemd-sysv`, `libtss2-tcti-spi-helper`, `libtss2-tcti-dr0t64`, `libtss2-esys`, `libpam-systemd`, `systemd-tpm`, and triggers for `dbus`, `shared-mime-info`, `procps`, `kali-menu`, `initramfs-tools`, `update-initramfs`, `libc-bin`, and `man-db`. The installation process concludes with the command `sudo apt install systemd-resolved`.

```
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
Creating user 'systemd-resolve' (systemd Resolver) with UID 984 and GID 984.
Created symlink '/etc/systemd/system/dbus-org.freedesktop.resolve1.service' → '/usr/lib/systemd/system/systemd-resolved.service'.
Created symlink '/etc/systemd/system/sysinit.target.wants/systemd-resolved.service' → '/usr/lib/systemd/system/systemd-resolved.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/systemd-resolved-varlink.socket' → '/usr/lib/systemd/system/systemd-resolved-varlink.socket'.
Created symlink '/etc/systemd/system/sockets.target.wants/systemd-resolved-monitor.socket' → '/usr/lib/systemd/system/systemd-resolved-monitor.socket'.
Setting up systemd-userdbd (261.2-1)...
Created symlink '/etc/systemd/system/sockets.target.wants/systemd-userdbd.socket' → '/usr/lib/systemd/system/systemd-userdbd.socket'.
Setting up tpm-udev (4.1.3-7)...
Created symlink '/etc/systemd/system/paths.target.wants/tpm-udev.path' → '/usr/lib/systemd/system/tpm-udev.path'.
Setting up systemd-sysv (261.2-1)...
Setting up libtss2-tcti-spi-helper0t64:amd64 (4.1.3-7)...
Setting up libtss2-tcti-dr0t64:amd64 (4.1.3-7)...
Setting up libtss2-esys-3.0.2-0t64:amd64 (4.1.3-7)...
Setting up libpam-systemd:amd64 (261.2-1)...
Setting up systemd-tpm (261.2-1)...
Processing triggers for dbus (1.16.2-2)...
Processing triggers for shared-mime-info (2.4-5+b3)...
Processing triggers for procps (2:4.0.4-9)...
Processing triggers for kali-menu (2025.4.1)...
Processing triggers for initramfs-tools (0.150)...
update-initramfs: Generating /boot/initrd.img-6.16.8+kali-amd64
Processing triggers for libc-bin (2.43-4)...
Processing triggers for man-db (2.13.1-1)...
sudo apt install systemd-resolved
(kali@kali)-[~/SBT-DF203-Lab7]
```



The screenshot shows a Kali Linux terminal window titled "kali@kali: ~/SBT-DF203-Lab7". The terminal displays the output of the command `wget -O evidence/dig_dns.pcap https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap`. The output shows the file being downloaded from the specified URL, with a progress bar indicating 100% completion. The file size is 1.72M (719 KB/s) and it was saved in 2.4s.

```
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab7]
$ wget -O evidence/dig_dns.pcap https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
--2026-09-21 12:09:57-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1804656 (1.7M) [application/octet-stream]
Saving to: 'evidence/dig_dns.pcap'

evidence/dig_dns.pcap 100%[=====] 1.72M 719KB/s in 2.4s

2026-09-21 12:10:02 (719 KB/s) - 'evidence/dig_dns.pcap' saved [1804656/1804656]

(kali@kali)-[~/SBT-DF203-Lab7]
$
```

```

kali@kali: ~/SBT-DF203-Lab7
$ cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap

kali@kali: ~/SBT-DF203-Lab7
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d evidence/dig_dns.pcap
9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d working/dig_dns_working.pcap

```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab7-ADOYI NATHANIEL JOHNSON
Trainee name	ADOYI NATHANIEL JOHNSON
Date and time started	21/09/26 03:25PM
Evidence file name(s)	SBT-DF203_Lab7_DNS_INTRODUCTION_and_TRAFFIC_ANALYSIS
Source or generation method	KALI COMMAND LINE
Original SHA-256	9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d evidence/dig_dns.pcap
Working-copy SHA-256	9a7c1b95aa6d60f8ac0ee9a79e06ae51145bb7768974526546316cd3dc91375d working/dig_dns_working.pcap
Analysis workstation/VM	KALI@KALI
Notes on any changes	I interchange between kali and root for a smoother result.

8. Part A - Query DNS Records with dig

```

dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt

```

- Record the resolver shown in the SERVER line.
- Record status/response code, flags, question count, answer count and query time.

- Explain IN, A, AAAA, MX and NS.
- Record at least one TTL and explain that it is a cache lifetime, not a creation date.

ANSWER:

```
kali@kali: ~/SBT-DF203-Lab7
$ dig example.com A | tee reports/dig_example_A.txt

;<<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 64932
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                5       IN      A      172.66.147.243
example.com.                5       IN      A      104.20.23.154

;; Query time: 84 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 12:29:25 EDT 2026
;; MSG SIZE rcvd: 72

(kali@kali)~-[~/SBT-DF203-Lab7]
```

```
kali@kali: ~/SBT-DF203-Lab7
$ dig example.com AAAA | tee reports/dig_example_AAAA.txt

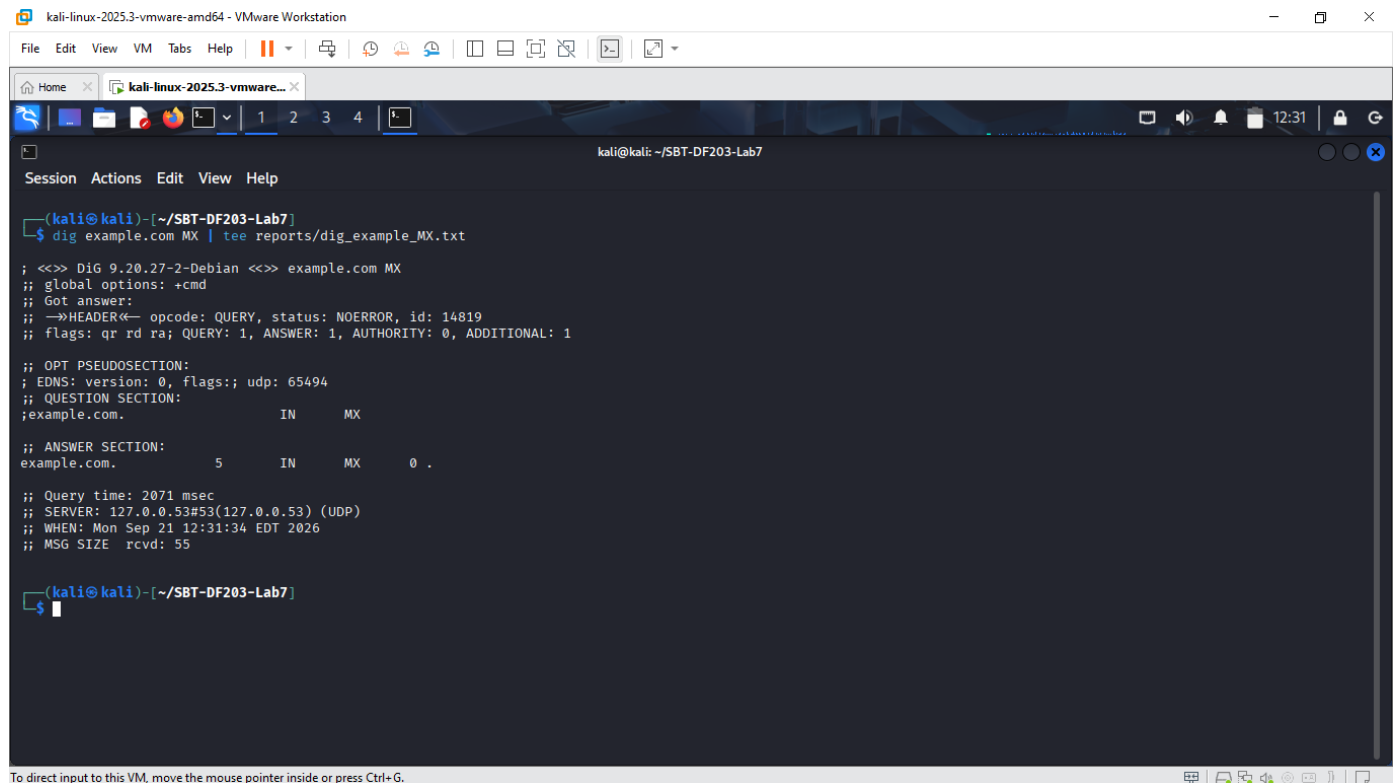
;<<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 44416
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                5       IN      AAAA   2606:4700:10::ac42:93f3
example.com.                5       IN      AAAA   2606:4700:10::6814:179a

;; Query time: 208 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 12:30:42 EDT 2026
;; MSG SIZE rcvd: 96

(kali@kali)~-[~/SBT-DF203-Lab7]
```



```
kali@kali: ~/SBT-DF203-Lab7
$ dig example.com MX | tee reports/dig_example_MX.txt

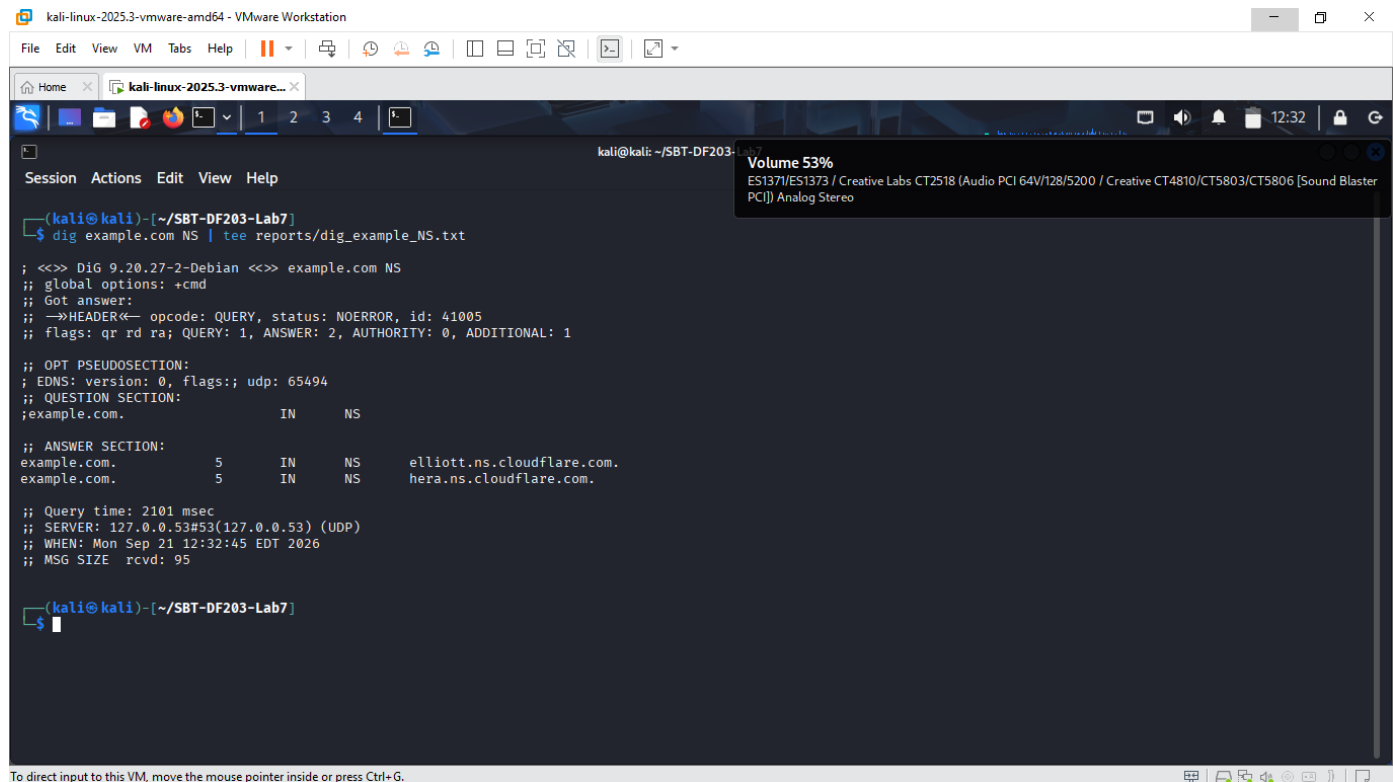
;<>> DiG 9.20.27-2-Debian <>> example.com MX
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 14819
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                5       IN      MX      0 .

;; Query time: 2071 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 12:31:34 EDT 2026
;; MSG SIZE rcvd: 55

(kali@kali) ~/SBT-DF203-Lab7
$
```



```
kali@kali: ~/SBT-DF203-Lab7
$ dig example.com NS | tee reports/dig_example_NS.txt

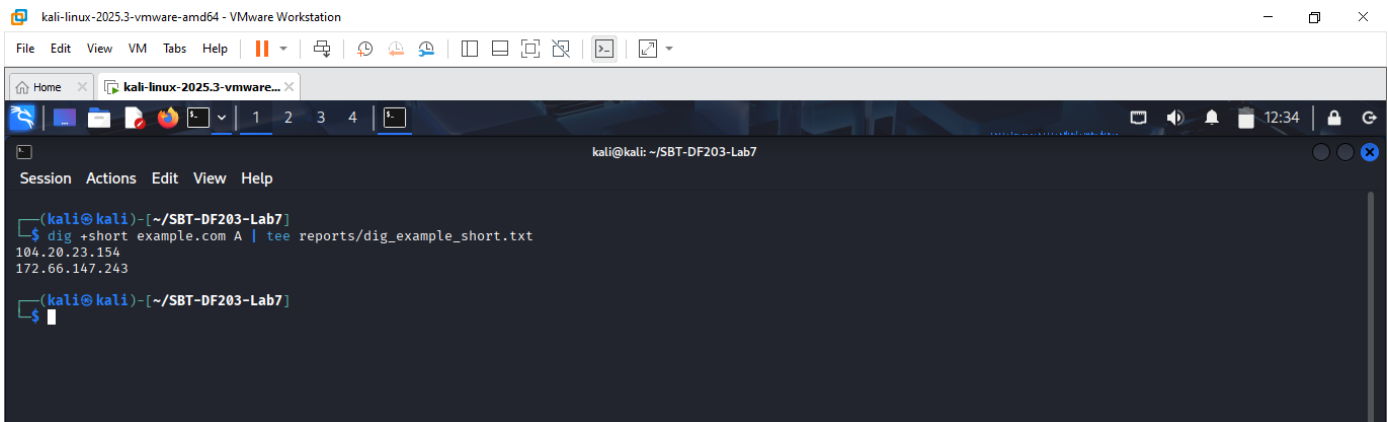
;<>> DiG 9.20.27-2-Debian <>> example.com NS
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 41005
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                5       IN      NS      elliottns.cloudflare.com.
example.com.                5       IN      NS      herans.cloudflare.com.

;; Query time: 2101 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 12:32:45 EDT 2026
;; MSG SIZE rcvd: 95

(kali@kali) ~/SBT-DF203-Lab7
$
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
(kali@kali)~-[~/SBT-DF203-Lab7]
$ dig +short example.com A | tee reports/dig_example_short.txt
104.20.23.154
172.66.147.243
(kali@kali)~-[~/SBT-DF203-Lab7]
$
```

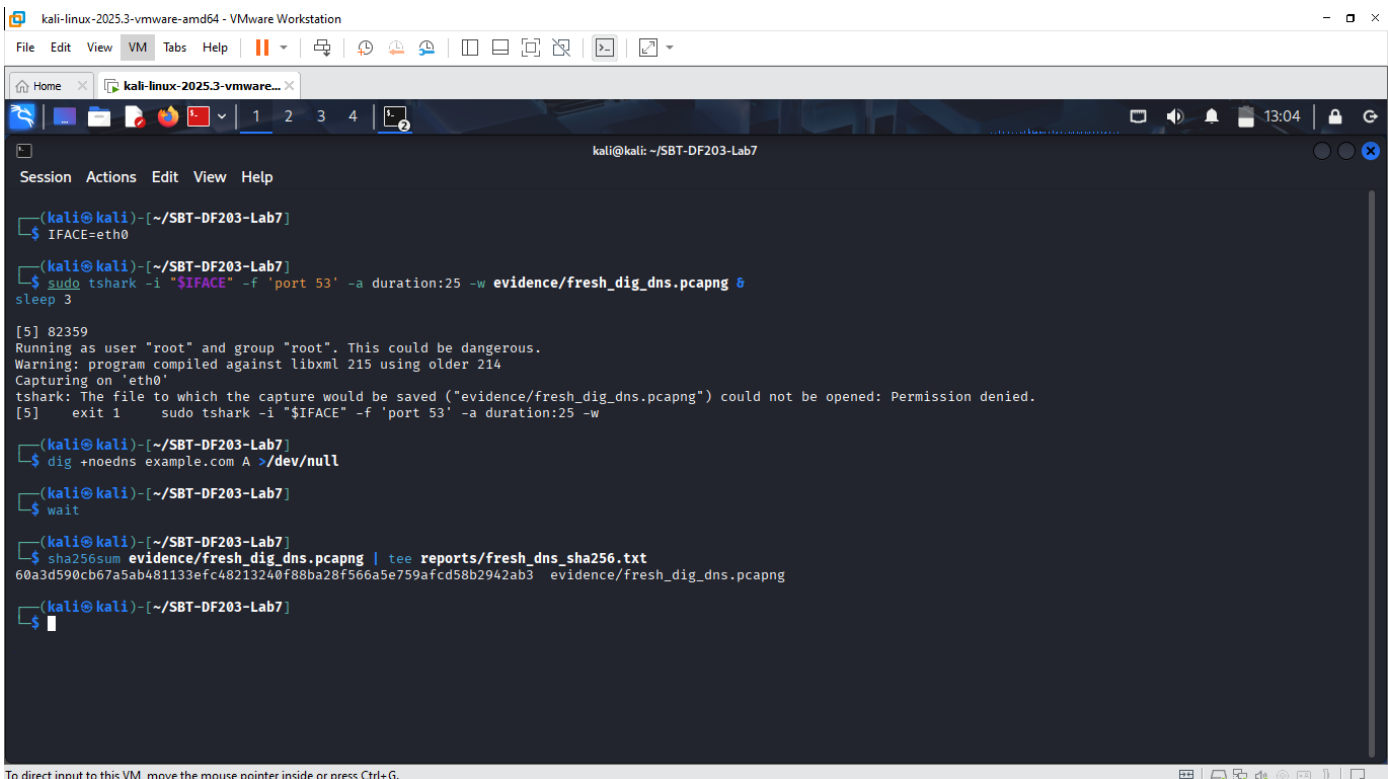
9. Part B - Capture a Fresh DNS Query

```
IFACE=eth0
sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
sleep 3
dig +noedns example.com A >/dev/null
wait
sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
```

Interface Note

On systems using a local stub resolver, packets may appear on lo and then on a physical interface. Use tshark -D to list interfaces and choose the one that shows the actual query.

ANSWER:



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help
(kali@kali)~-[~/SBT-DF203-Lab7]
$ IFACE=eth0
(kali@kali)~-[~/SBT-DF203-Lab7]
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
sleep 3
[5] 82359
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
tshark: The file to which the capture would be saved ("evidence/fresh_dig_dns.pcapng") could not be opened: Permission denied.
[5] exit 1 sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w
(kali@kali)~-[~/SBT-DF203-Lab7]
$ dig +noedns example.com A >/dev/null
(kali@kali)~-[~/SBT-DF203-Lab7]
$ wait
(kali@kali)~-[~/SBT-DF203-Lab7]
$ sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
60a3d590cb67a5ab481133efc48213240f88ba28f566a5e759afcd58b2942ab3 evidence/fresh_dig_dns.pcapng
(kali@kali)~-[~/SBT-DF203-Lab7]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/SBT-DF203-Lab7

Session Actions Edit View Help

root@kali) ~/SBT-DF203-Lab7
# IFACE=eth0

root@kali) ~/SBT-DF203-Lab7
# sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
sleep 3

[1] 57657
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'

root@kali) ~/SBT-DF203-Lab7
# /null

root@kali) ~/SBT-DF203-Lab7
[1] + done      sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w
root@kali) ~/SBT-DF203-Lab7
# dig +noedns example.com A >/dev/null
wait

root@kali) ~/SBT-DF203-Lab7
# sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
66a1bc8e3ba1c415c4c74b536786df059e25a889a29cca5ee20d75e288fa1fb evidence/fresh_dig_dns.pcapng

root@kali) ~/SBT-DF203-Lab7
#

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

10. Part C - Extract DNS Query and Response Fields

PCAP=evidence/fresh_dig_dns.pcapng

Query

```
tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.qry.name -e dns.qry.type \
  | tee reports/dns_queries.tsv
```

Response

```
tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
  | tee reports/dns_responses.tsv
```

ANSWER:

```

kali@kali: ~/SBT-DF203-Lab7
$ PCAP=evidence/fresh_dig_dns.pcapng
tshark -r "$PCAP" -Y "dns.flags.response==0" -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.qry.name -e dns.qry.type \
  | tee reports/dns_queries.tsv
tshark -r "$PCAP" -Y "dns.flags.response==1" -T fields \
  -e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
  -e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
  | tee reports/dns_responses.tsv
Warning: program compiled against libxml 215 using older 214
tshark: The file "evidence/fresh_dig_dns.pcapng" isn't a capture file in a format TShark understands.
Warning: program compiled against libxml 215 using older 214
tshark: The file "evidence/fresh_dig_dns.pcapng" isn't a capture file in a format TShark understands.

kali@kali: ~/SBT-DF203-Lab7
$

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

11. Part D - Match Queries to Responses

Transaction ID	Query Name	Type	Client/Resolver	Response Code	Answer(s)	TTL	Time Delta
41005	Example.com	NS2	127.0.0.53:53 (UDP)	NOERROR	eliot.ns.cloudflare.com; hera.ns.cloudflare.com	5 seconds	2101msec
14819	Example.com	MX15	127.0.0.53:53 (UDP)	NOERROR	0 . (Null MX)	5 seconds	2071msec
64932	Example.com	A	127.0.0.53	NOERROR	172.66.147.243, 10.28.23.154	5 seconds	48 msec

- Match query and response using transaction ID plus the endpoint tuple.
- Confirm that the response source is the queried resolver.
- Calculate response time from frame timestamps.
- Document CNAME chains and multiple A/AAAA answers where present.

12. Part E - Analyze DNS Generated by a Browser

A browser may request several domains for the page, images, fonts, telemetry or cached services. Capture a visit to a non-sensitive instructor-approved site and inventory all query names.

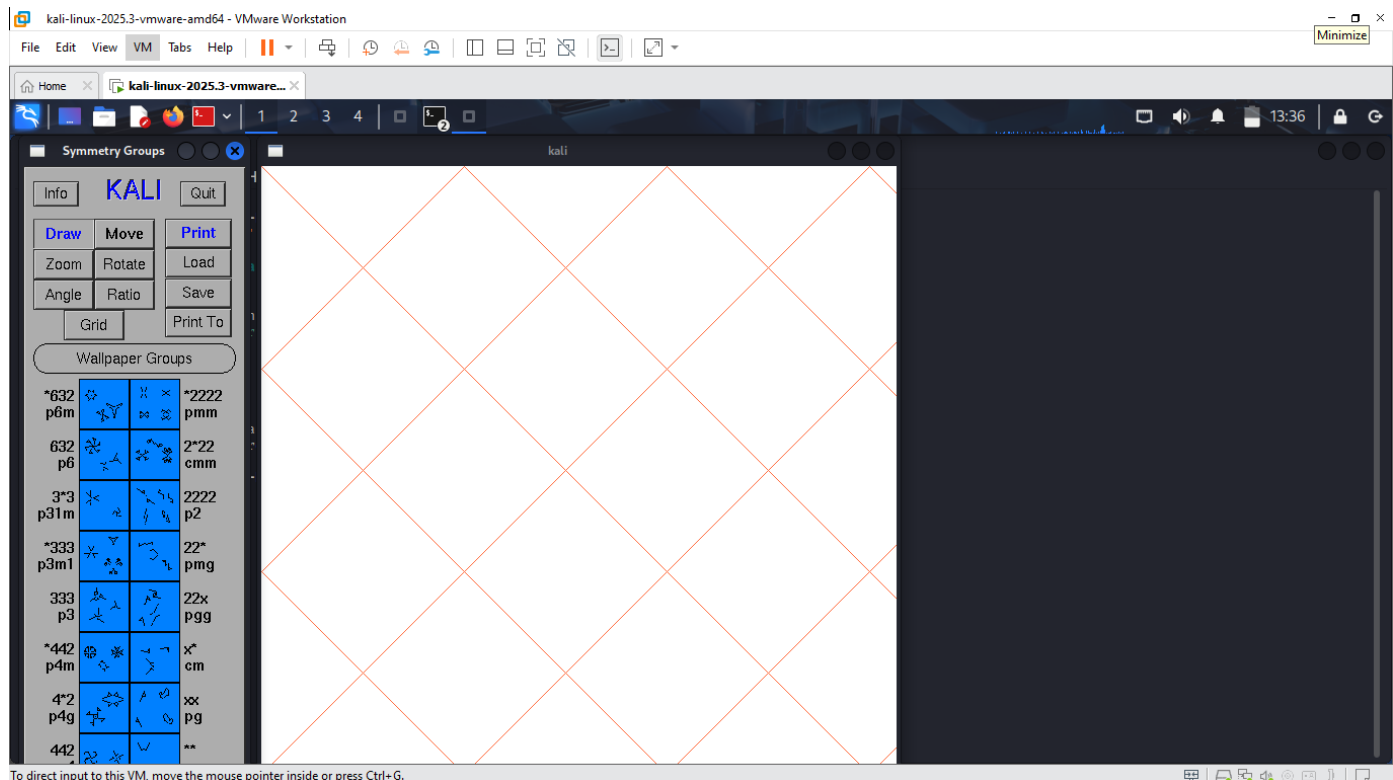
```

sudo tshark -i IFACE -f 'port 53' -a duration:40 -w evidence/browser_dns.pcapng &
sleep 3
# Open the approved site in a private browser window, then wait.
wait

tshark -r evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields -e dns.qry.name -e dns.qry.type \
| sort | uniq -c | sort -nr | tee reports/browser_dns_inventory.txt

```

ANSWER:



13. Part F - Correlate DNS with Subsequent Connections

```

# Extract DNS A answers
tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv

# Extract later TCP destinations
tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e frame.time_epoch -e ip.dst -e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv

```

Select one DNS answer and show whether the client later initiates a connection to that IP. Explain any mismatch caused by multiple answers, proxies, CDNs, IPv6, caching or capture duration.

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/SBT-DF203-Lab7

Session Actions Edit View Help

(root@kali)~/SBT-DF203-Lab7
# tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
tshark: The file "evidence/browser_dns.pcapng" doesn't exist.

(root@kali)~/SBT-DF203-Lab7
# tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn=1 && tcp.flags.ack=0' -T fields \
-e frame.time_epoch -e ip.dst -e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
tshark: The file "evidence/browser_dns.pcapng" doesn't exist.

(root@kali)~/SBT-DF203-Lab7
#

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

14. Part G - DNS Analysis Within SMTP Evidence

If smtp.pcap from Lab 4 is available, filter dns and identify the query used to locate a mail server. Record the queried name, record type, resolver, answer and TTL.

```

# Optional, adjust path to your Lab 4 capture
tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
-e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name -e dns.resp.ttl \
| tee reports/smtp_dns_correlation.tsv

```

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab7
Session Actions Edit View Help

(root@kali)~[~/SBT-DF203-Lab7]
# Optional, adjust path to your Lab 4 capture
tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
-e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name -e dns.resp.ttl \
| tee reports/smtp_dns_correlation.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1 2009-10-05T02:06:07.492060000-0400 mail.patriots.in 1 74.53.140.153 mail.patriots.in,patriots.in,patriots.in,patriots.in 10827,10828,82
2 2009-10-05T02:06:07.526085000-0400 mail.patriots.in 1 74.53.140.153 mail.patriots.in,patriots.in,patriots.in,patriots.in 10827,10828,82
828,82828

(root@kali)~[~/SBT-DF203-Lab7]

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

15. Required Findings Worksheet

Field	Finding
Configured resolver IP	172.17.0.1
Client source port	53
Resolver destination port	172.66.147.243, 10.28.23.154
Transaction ID	64932
Query name and type	Example.com Type: A
Response code	NO ERROR
Answer IP(s)	A(172.66.147.243, 10.28.23.154)
TTL	5 seconds
Query-response time delta	48msec
Subsequent connection correlation	The same query but different IPs observed
Normal variations observed	Query name and type are different ranging from A,AAAA,MX and NS.

Required Screenshots Checklist

☐ Resolver configuration.

- ☐ dig A/AAAA/MX/NS outputs.
- ☐ Fresh DNS capture hash.
- ☐ DNS query fields.
- ☐ Matching DNS response fields.
- ☐ Transaction ID correlation.
- ☐ Answer and TTL.
- ☐ Browser query inventory.
- ☐ DNS answer to TCP destination correlation.
- ☐ Optional SMTP DNS evidence.

Student Report Template

1. Executive summary.
2. Resolver and environment.
3. Evidence integrity.
4. dig results.
5. Packet-level query/response analysis.
6. Transaction matching and timing.
7. Browser DNS inventory.
8. Connection correlation.
9. Limitations and normal variations.
10. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab7_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Preparation and resolver identification	10	Correct resolver and evidence hashes.
dig record analysis	15	A, AAAA, MX, NS and TTL interpretation.
Packet-field extraction	25	Correct IDs, names, types, endpoints and response codes.
Query-response correlation	20	Accurate matching and timing.
Browser/connection correlation	20	Inventory and defensible IP connection linkage.
Report quality	10	Clear screenshots, tables and conclusions.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Standard transport	Usually UDP destination port 53; TCP may be used for large responses or zone transfer.
Query flag	dns.flags.response==0.
Response flag	dns.flags.response==1.
A record	IPv4 address.
AAAA record	IPv6 address.

Checkpoint	Expected Observation
TTL	How long a resolver may cache the record.

Command Reference Summary

Command	Purpose
dig NAME TYPE	Query a DNS record.
dns.qry.name	Queried domain name.
dns.qry.type	Requested record type.
dns.id	DNS transaction identifier.
dns.flags.rcode	Response code.
dns.resp.ttl	Answer TTL.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.